



COVID-19 Vaccination Trends and Global Health Inequality

How do vaccination trends differ across countries, and what do they reveal about global health inequality?

Eva Štětínová
Ayushman Mandhotra

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Overview

This report analyses global COVID-19 vaccination trends using the Our World in Data COVID-19 vaccination dataset (Mathieu et al., 2021), covering 222 countries and territories between December 2, 2020, and August 28, 2021. Using descriptive statistics, principal component analysis, k-means clustering, and ARIMA time series forecasting, the analysis quantifies differences in vaccination coverage, rollout speed, and data-reporting completeness across countries and World Bank income groups.

Only 28.2% of countries reached 50% full vaccination coverage during the observation period, and coverage differed by more than 50 percentage points between the highest- and lowest-income groups. A forecasting comparison across Spain, India, and Bangladesh further shows that the reliability of short-term vaccination forecasts is itself unequally distributed once appropriately scaled for each country's own coverage level and extending the inequality finding beyond outcomes and into the data infrastructure used to monitor them.

Keywords: *COVID-19, vaccination, global health inequality, principal component analysis, k-means clustering, ARIMA forecasting*



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1. Introduction

The COVID-19 pandemic presented an unprecedented challenge to global public health systems. The rapid development and subsequent worldwide deployment of COVID-19 vaccines in 2020 and 2021 represented a landmark achievement in medical science, but the distribution and administration of those vaccines across countries was deeply uneven. While early commentary on vaccine inequity relied largely on aggregate, country-level snapshots, the daily, publicly reported data compiled by Our World in Data make it possible to examine not only how much of the world was vaccinated, but how quickly, how consistently, and how reliably that progress could be tracked and forecast in the first place.

This report presents a data-driven analysis of COVID-19 vaccination trends across 222 countries and territories during the first nine months of global vaccine deployment, from December 2020 to August 2021. The analysis is organized around a central research question: how do vaccination trends differ across countries, and what do they reveal about global health inequality?

2. Problem Definition

2.1 State of the Art

Vaccine inequity during the COVID-19 pandemic has been documented extensively in the scientific literature. Wouters et al. (2021) identified multiple structural barriers to equitable vaccine access, including financial constraints, manufacturing capacity, and geopolitical factors that advantaged wealthier countries in the first year of the rollout. The COVAX initiative, established to pool procurement and ensure more equitable global distribution, faced substantial and well-documented obstacles in reaching low-income countries during 2021. More broadly, Acharya and Porwal showed that pre-existing structural healthcare inequalities were strong predictors of differential outcomes during the pandemic, a pattern consistent with the vaccination-specific findings in this report.

Data availability is itself a dimension of health inequality. Countries with stronger health information systems have historically been able to report disease and health-programme data continuously, while countries with weaker administrative or digital infrastructure often report sporadically or not at all. This has direct implications for monitoring vaccination campaigns and for any statistical model including a forecasting model that depends on that reported data being complete and regularly spaced in time.

2.2 Problem Statement

Despite extensive documentation of vaccine inequity at the macro level, granular, reproducible, data-driven analyses that compare vaccination trajectories across countries using a single public dataset and that treat the reliability of the data and of any resulting forecast as part of the inequality story, not just the outcomes themselves remain comparatively rare. This report addresses four specific gaps:

- ☀ How large is the quantitative gap in vaccination coverage between high-income and low-income countries, and does it widen or narrow over the observation period?
- ☀ Do differences manifest only in final coverage, or also in the speed of rollout?
- ☀ Does data missingness correlate with a country's apparent vaccination performance, and what does that imply for monitoring?
- ☀ Can time series forecasting models reliably project vaccination progress, and is forecast reliability itself unequal across countries and if so, is that inequality still present once differences in absolute coverage level are accounted for?

3. Objectives

The primary objective of this study is to characterize global COVID-19 vaccination inequality through a reproducible, quantitative analysis of a single publicly available dataset. The specific analytical objectives are:

- ☀ To describe the distribution of vaccination coverage across countries using descriptive statistics and a diverse set of visualization techniques.
- ☀ To compare vaccination rollout speed by measuring the time required for countries to reach 50% full vaccination coverage.
- ☀ To analyse income-group and continental differences in vaccination trajectories, and to quantify how reporting completeness itself varies across countries.
- ☀ To apply unsupervised learning methods to identify country groupings by vaccination profile, with the number of clusters justified empirically rather than assumed.
- ☀ To evaluate the predictive reliability of ARIMA time series models for vaccination forecasting across countries at different coverage levels, and to test rather than assume whether that reliability tracks income level.

4. Methods

4.1 Dataset

The analysis uses the Our World in Data COVID-19 vaccination dataset with daily observations for 234 locations, of which 222 are individual countries or territories and 12 are continent- or income-group aggregates (World, Europe, European Union, Asia, Africa, North America, South America, Oceania, and the four World Bank income groups). The country-level data span December 2, 2020, to August 28, 2021, and provide 14 variables per observation, including total vaccinations, people vaccinated, people fully vaccinated, and booster doses, each available as both an absolute count and a rate per 100 inhabitants.

4.2 Data Cleaning

The raw file was loaded directly with missing values already encoded as empty cells. Percentage-type columns were coerced to numeric type, and date strings were parsed as calendar dates. Regional and income-group aggregate rows were separated from individual country rows. Aggregates were retained for the income-group comparison but excluded from every country-level analysis to avoid double-counting, leaving 222 individual countries and territories, of which 216 have at least one non-missing full-vaccination record.

4.3 Descriptive Statistics and Visualisation

Descriptive statistics were computed for the primary outcome variable, people fully vaccinated per 100 inhabitants, including the distribution's extremes and its variation across countries. A cross-sectional snapshot was constructed by retaining each country's own latest available observation; because these "latest" dates range from April to August 2021 across countries, comparisons based on this snapshot are noted as a limitation throughout.

All visualizations were produced using **ggplot2** with a consistent colour palette adjusted for colorblindness, together with several extension packages **ggbreak** for axis-break charts, **ggdist** and **ggbeeswarm** for the raincloud plot, and **ggcorrplot** for the correlation heat map.

4.4 Rollout Speed Analysis

To quantify differences in vaccination speed independently of final coverage, each country was classified according to whether it ever reached 50% full vaccination coverage during the observation period. For countries that did, the number of days between their first recorded vaccination observation and the date that threshold was reached was calculated as a proxy for rollout speed.

4.5 Principal Component Analysis (PCA)

PCA was applied to four vaccination indicators such as total vaccinations, people vaccinated, people fully vaccinated (all per hundred), and daily vaccinations per million after standardization to zero mean and unit variance. PCA reduces these correlated indicators to a small number of orthogonal components, each capturing a share of the total variance; the first two components were retained for visualization.

4.6 K-means Clustering

K-means clustering was applied to the same standardized indicators to identify distinct country groupings by vaccination profile. Rather than assuming three clusters from theory alone, the number of clusters was first assessed using the within-cluster sum-of-squares ("elbow") method and the average silhouette width, both computed with the **factoextra** package. $k = 3$ was retained on the basis of these diagnostics together with the theoretically expected high/moderate/low coverage segmentation. A fixed random seed was set prior to model fitting, in accordance with the minimum standard of reproducibility. Because k-means assigns cluster numbers arbitrarily, the three resulting clusters were labelled according to each cluster's own computed mean coverage.

4.7 ARIMA Time Series Forecasting

Autoregressive Integrated Moving Average models were fitted to the daily vaccination coverage series of selected countries using the ***auto.arima()*** function from the **forecast** package, which selects its own differencing order via internal unit-root tests rather than a fixed, assumed order. A weekly seasonal frequency was specified to match the data's reporting cycle. Because not every country reports on every calendar day, a verified gap of 10 multi-day jumps in India's series and 32 in Bangladesh's, versus none in Germany's each country's series was first re-expressed on a complete daily calendar with missing days linearly interpolated before model fitting, so that the weekly seasonal index in the model corresponds to actual calendar days rather than to raw row position.

Germany's series was modelled first as a single-country case study, with residual diagnostics, an ETS benchmark model, and an 80/20 train-test split used to validate the modelling approach. The same modelling strategy was then extended, unchanged, to a three-country comparison: Spain, India, and Bangladesh, chosen to span a high-income, a middle-income, and a lower-middle-income country. An Augmented Dickey-Fuller test was used alongside each fitted model's own selected order to characterize the stationarity properties of each country's series. Forecast reliability was then summarized with two complementary metrics: the mean width of each country's 95% prediction interval over a 30-day horizon in absolute percentage points, and the same width expressed as a share of that country's own forecast level, to separate genuine differences in reliability from the mechanical effect of a country's coverage being closer to, or further from, saturation.

5. Results

5.1 Descriptive Overview

Figure 1 compares five countries Spain, Germany, Czechia, Brazil, and India chosen to span high-income European states and two larger middle- and lower-middle-income countries. Spain, Germany, and Czechia show considerably faster rollout trajectories than Brazil and India; India remains clearly below the 50% threshold throughout the observation window.

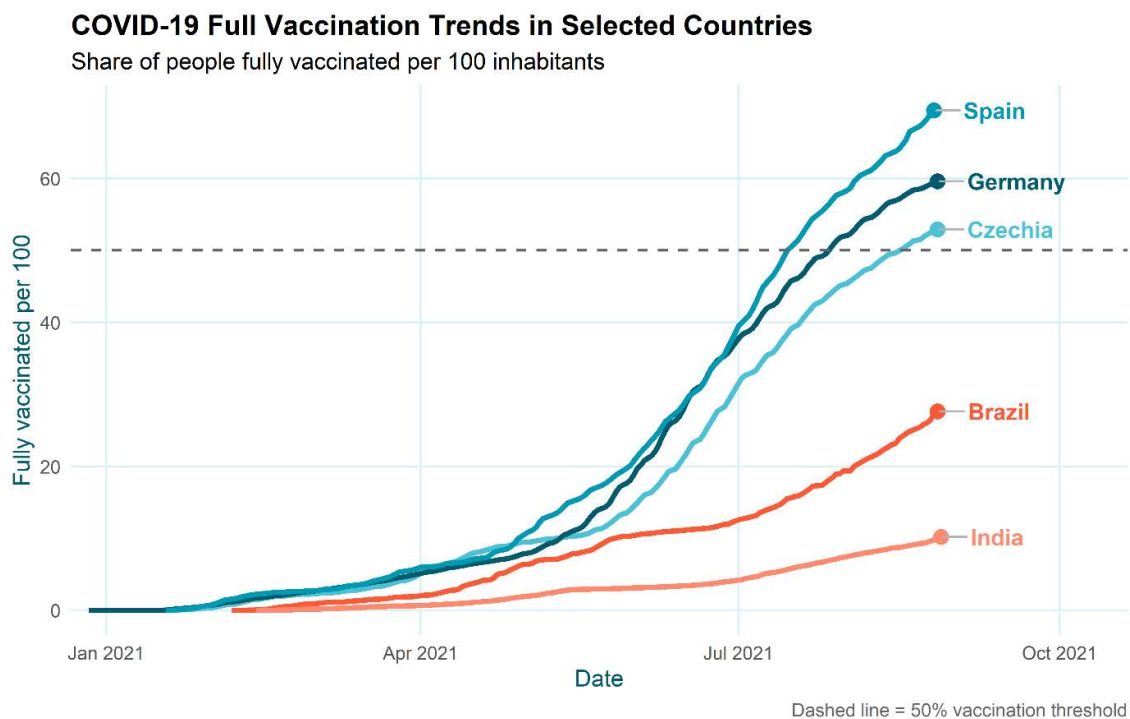


Figure 1. COVID-19 full vaccination trends in five selected countries. The dashed line marks the 50% threshold.

Figures 2 and 3 extend this comparison to the full set of 216 countries with available data. The highest-coverage country, Gibraltar, reported 116.4 fully vaccinated per 100 inhabitants at its latest observation a figure above 100 because it reflects vaccination of non-resident cross-border workers, not an error in the data. The lowest, the Democratic Republic of the Congo, reported 0.01, a ratio of roughly 11,640-fold between the two extremes.

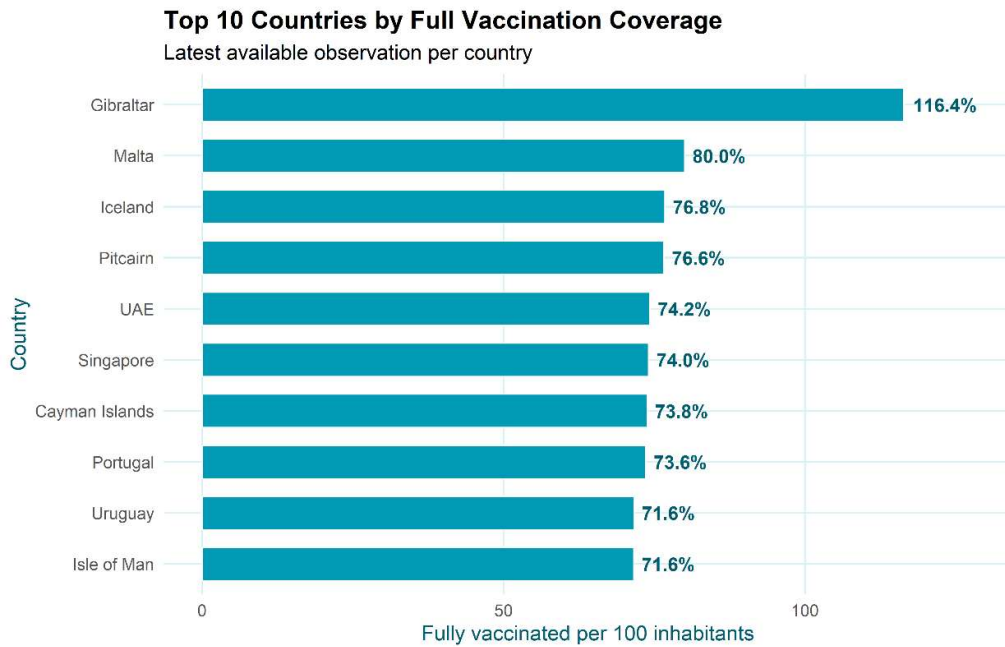


Figure 2. Top 10 countries by full vaccination coverage (latest available observation per country).

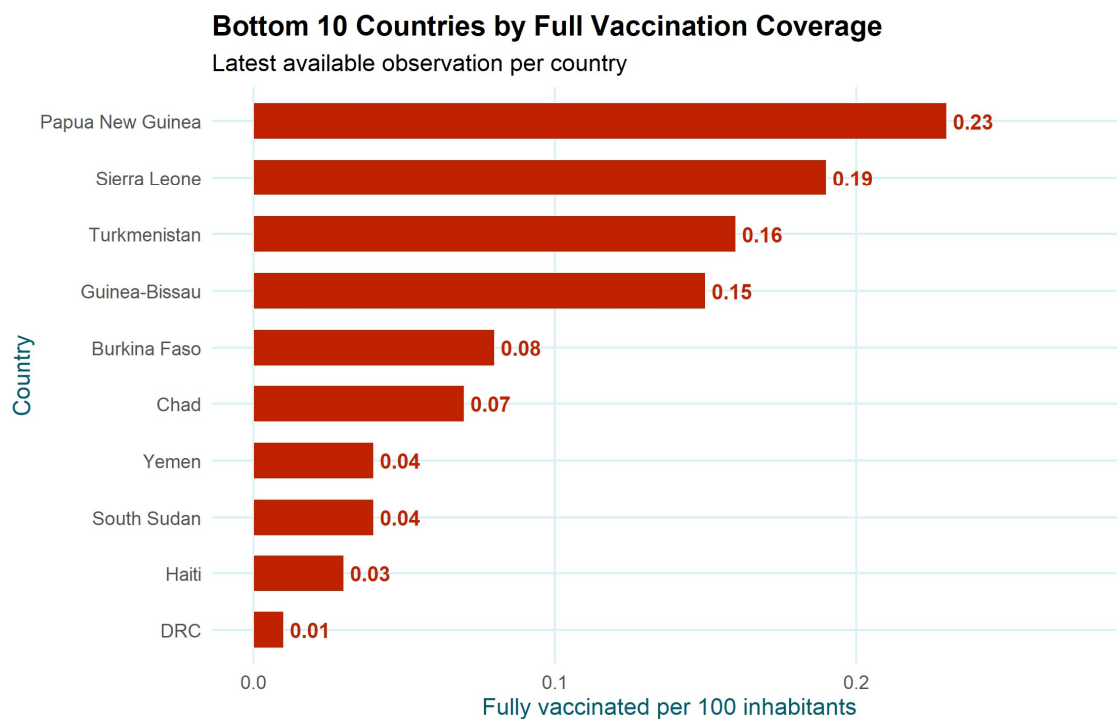
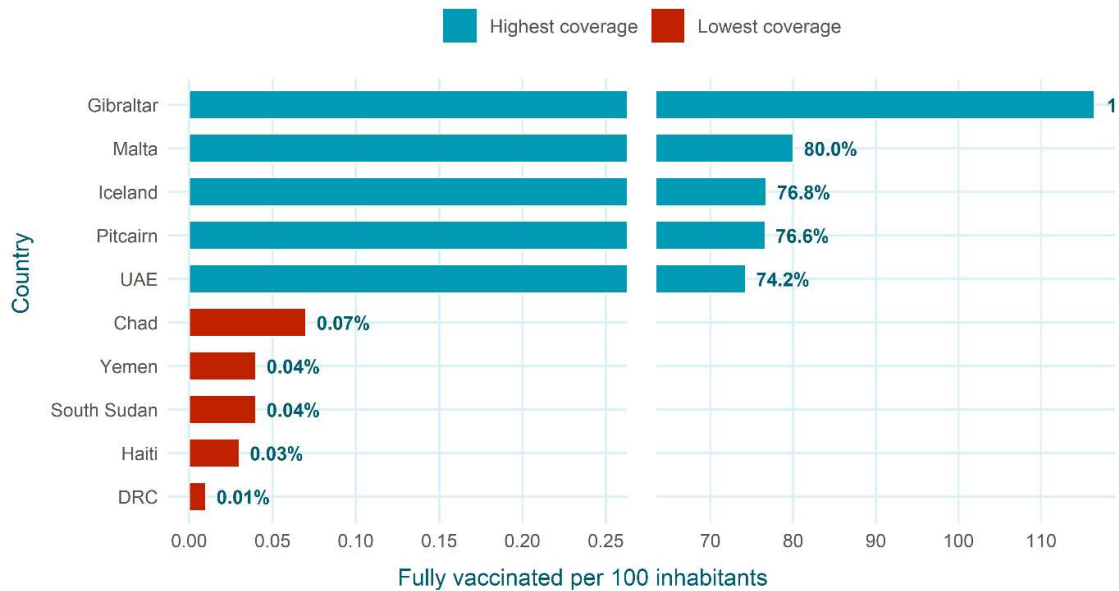


Figure 3. Bottom 10 countries by full vaccination coverage (latest available observation per country).

Because this gap spans more than three orders of magnitude, Figure 4 places the five highest- and five lowest-coverage countries on a single chart using an axis break.

Vaccination Coverage: Highest and Lowest Countries

Axis break separates the two ranges (gap between 0.25 and 66)



Source: Our World in Data COVID-19 vaccination dataset (Mathieu et al., 2021)

Figure 4. Highest- and lowest-coverage countries shown together on a broken axis.

5.2 Rollout Speed

Of the 216 countries with any reported full-vaccination data, only 61 (28.2%) reached the 50% coverage threshold at any point during the observation window. Figure 5 lists the ten fastest of those 61 countries by name: Pitcairn, Falkland Islands, Bhutan, Cook Islands, Nauru, Saint Helena, Gibraltar, Mongolia, China, and Qatar.

Among the 61 countries that reached the threshold, the median time was 160 days and the mean 139 days (Figure 6); the mean falling below the median indicates a small number of unusually rapid campaigns pulling the average down, consistent with the right-skew visible in the histogram.

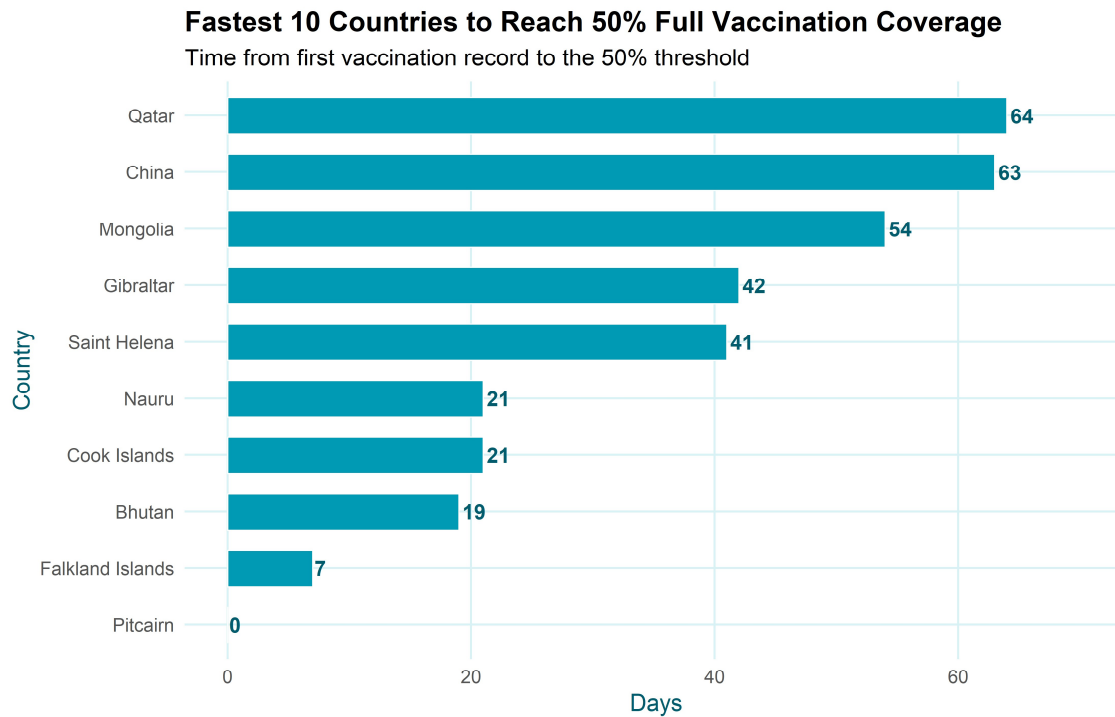


Figure 5. Fastest 10 countries to reach 50% full vaccination coverage.

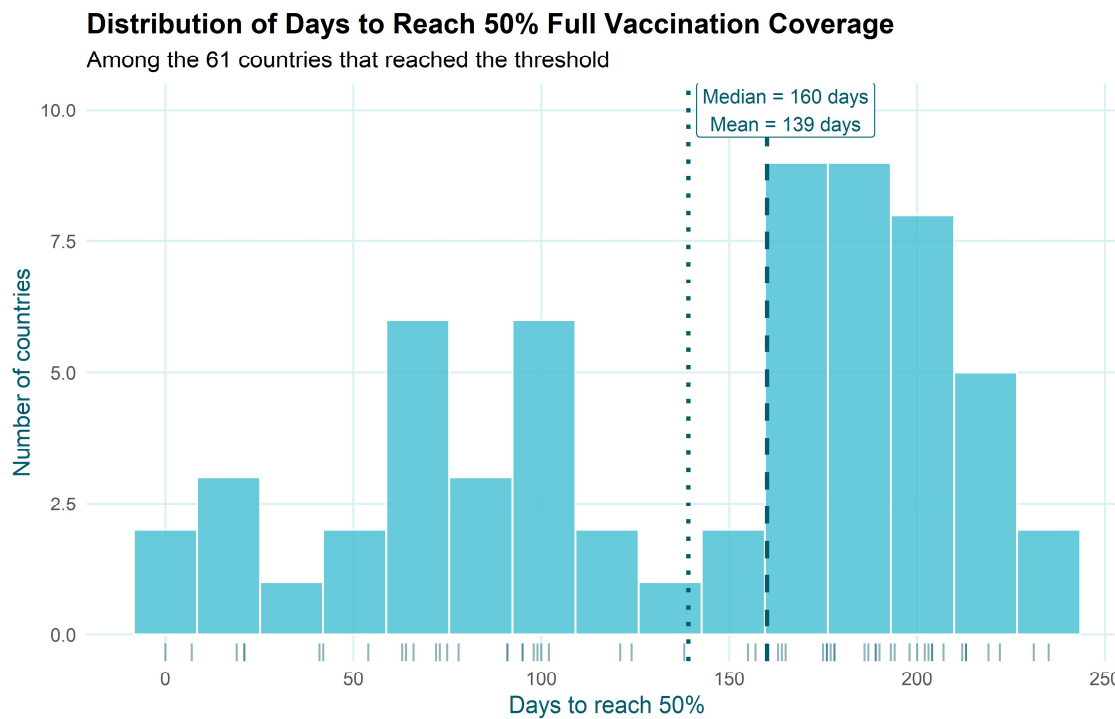


Figure 6. Distribution of days required to reach 50% full vaccination coverage, among the 61 countries that achieved it.

5.3 Missing Data

Thirty-six of the 216 countries have 95% or more of their full-vaccination records missing (Figure 8). High missingness does not necessarily indicate the absence of a vaccination campaign: Tokelau is 100% missing from this dataset yet is reported elsewhere to have achieved over 96% coverage of its eligible population.

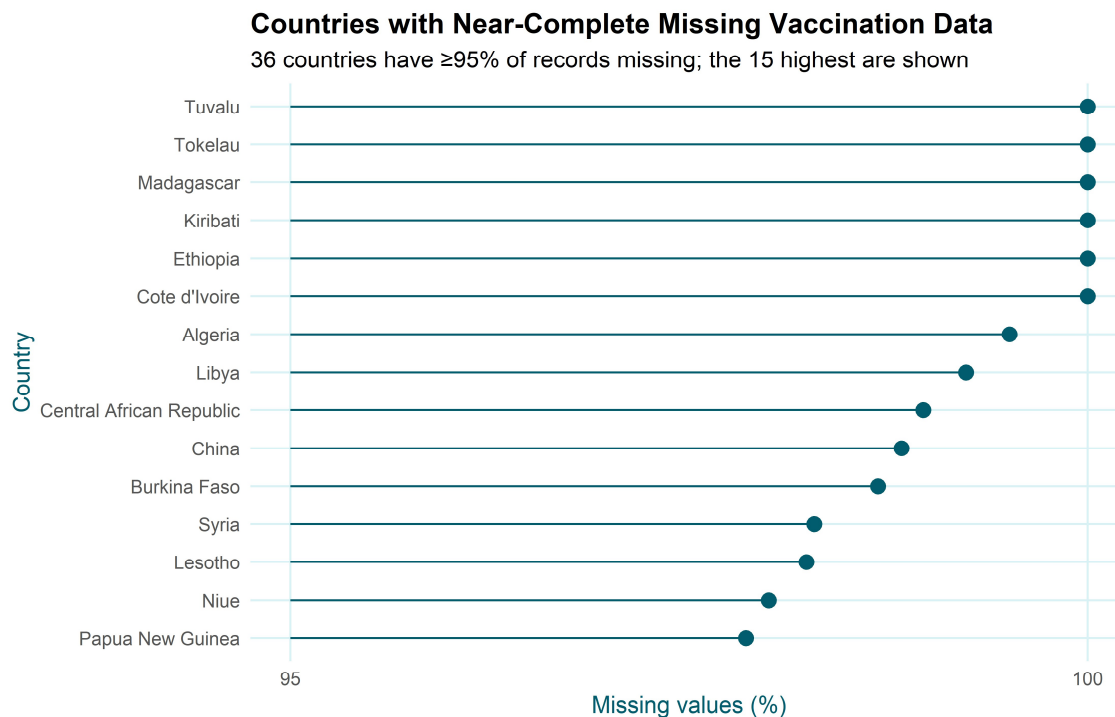


Figure 8. Countries with near-complete missing vaccination data ($\geq 95\%$ of records missing; 15 highest shown).

5.4 Geographic Patterns

Figures 9 and 10 present the same underlying inequality from two complementary geographic angles: overall coverage, and the speed at which that coverage was reached. High coverage in Figure 9 is concentrated in Europe, parts of the Americas, and isolated high-income states in Asia and Oceania; the African continent shows predominantly low shading or grey (no data). Figure 10 shows the same pattern for rollout speed, though grey on this map conflates two different things: the majority of countries that never reached the threshold at all and no data available.

Full Vaccination Coverage by Country

Latest available observation per country — grey = no data reported

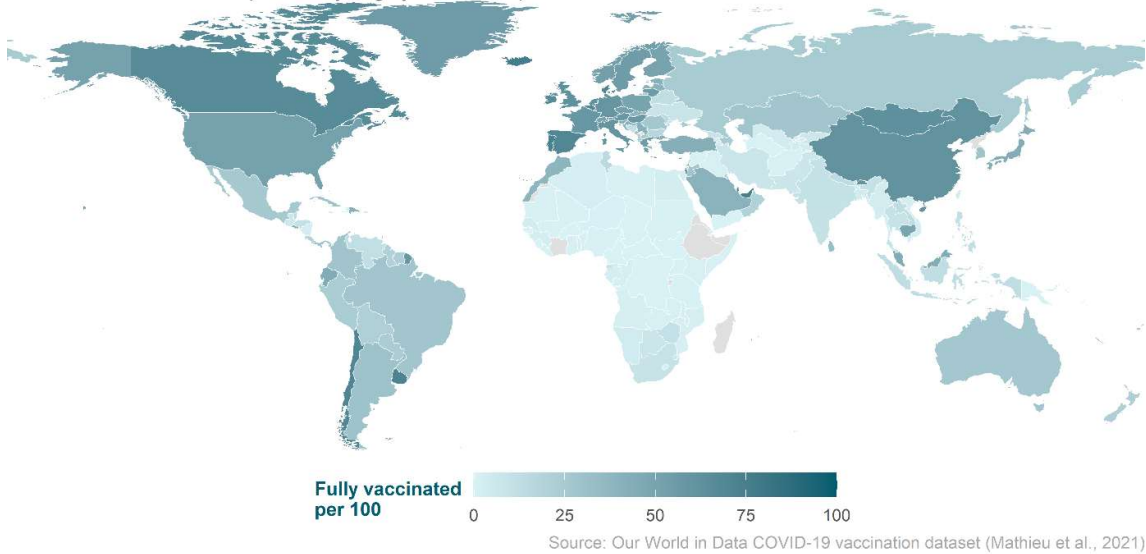


Figure 9. World map of full vaccination coverage by country (latest available observation).

Speed of Vaccination Rollout by Country

Days from first recorded vaccination to 50% coverage — grey = threshold never reached, or no data

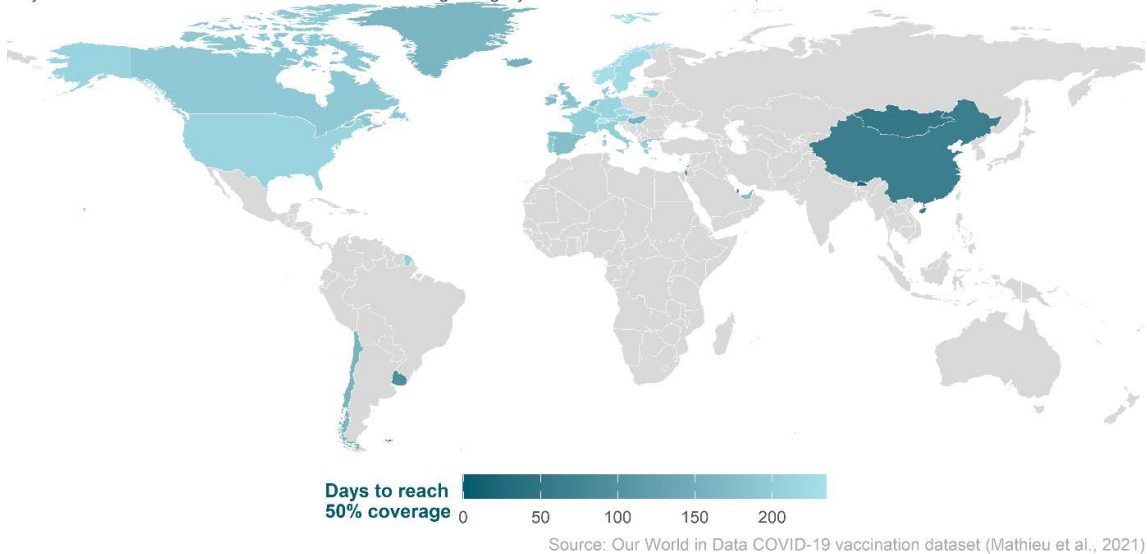
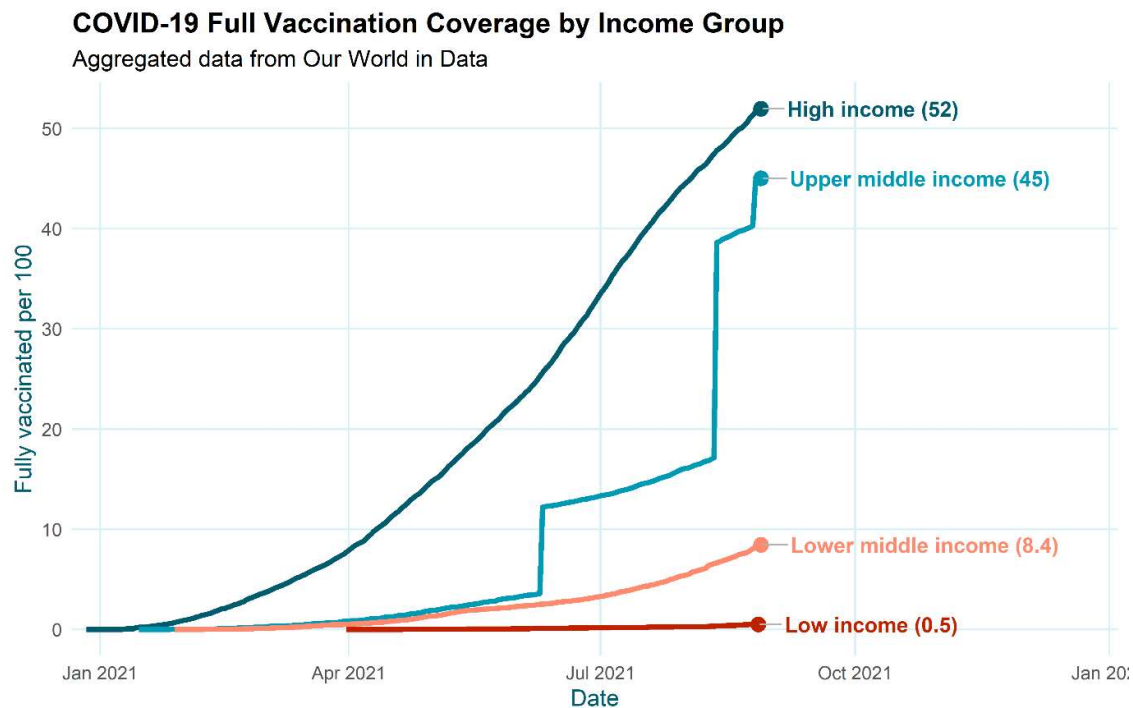


Figure 10. World map of vaccination rollout speed by country (days to reach 50% coverage).

5.5 Income-Group and Continental Inequality

The income-group time series (Figure 11) shows a gap between income groups that widens continuously across the observation period, accelerating from roughly April 2021 onward. By August 2021, high-income countries had reached approximately 52 fully vaccinated per 100 inhabitants, versus roughly 0.5 for low-income countries.

Figure 11. COVID-19 full vaccination coverage by World Bank income group over time.



The continent-level comparison (Figure 12) confirms this pattern at finer geographic resolution. One outlier, Gibraltar, was excluded from this specific figure because its coverage above 100% would compress the rest of the distribution. Africa shows the lowest median coverage and a right-skewed distribution; Europe the highest median and the narrowest spread; Asia and the Americas both show wide within-continent variation, indicating that continental groupings alone understate the heterogeneity driven by income and infrastructure differences.

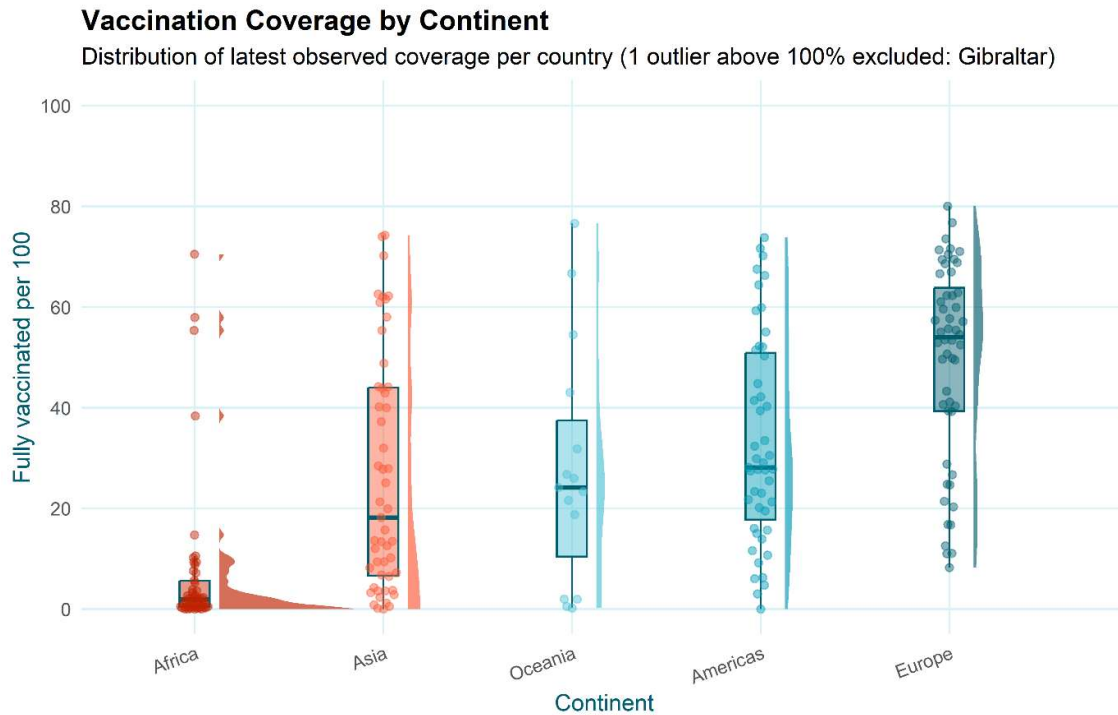


Figure 12. Distribution of vaccination coverage by continent (latest observation per country; one outlier above 100% excluded).

5.6 Correlation Analysis

Vaccination indicators show strong positive correlations among the three cumulative measures ($r = 0.94\text{--}0.98$): countries that lead on one lead on all of them. Daily vaccinations per million, by contrast, correlates only weakly to moderately with cumulative progress ($r = 0.16\text{--}0.40$) a country's most recent daily activity says little about its overall campaign success, since completed rollouts taper off their daily volumes while early-stage countries can show high daily rates off a still-low base.

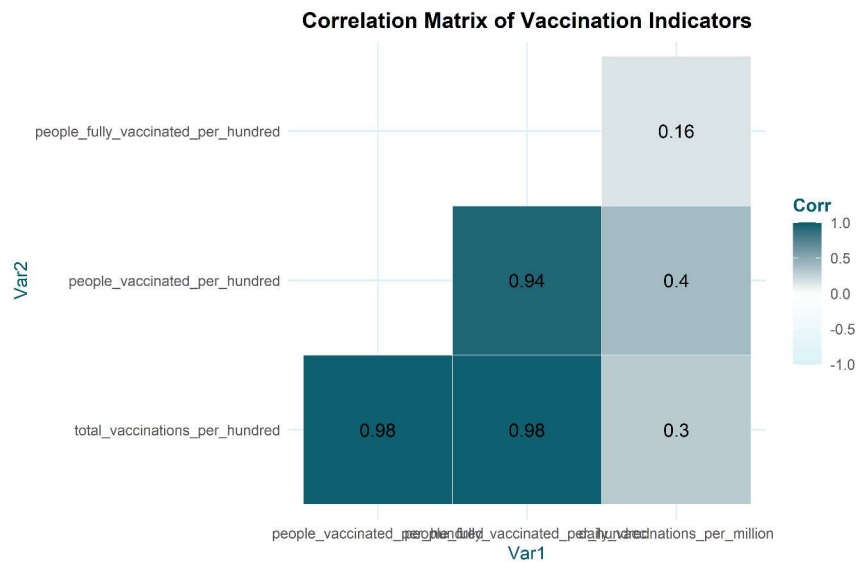


Figure 13. Correlation matrix of vaccination indicators (latest observation per country).

5.7 PCA and K-means Clustering

Principal component analysis was computed on 209 of the 216 countries with complete data across all four indicators. The first component explained 76.5% of total variance and the second 22.8%, together accounting for 99.3% matching, rather than contradicting, the figures reported for the same computation elsewhere in this report. PC1 separates countries primarily by overall vaccination coverage, with lower-income African and Asian countries clustering at one end and European countries at the other (Figure 14).

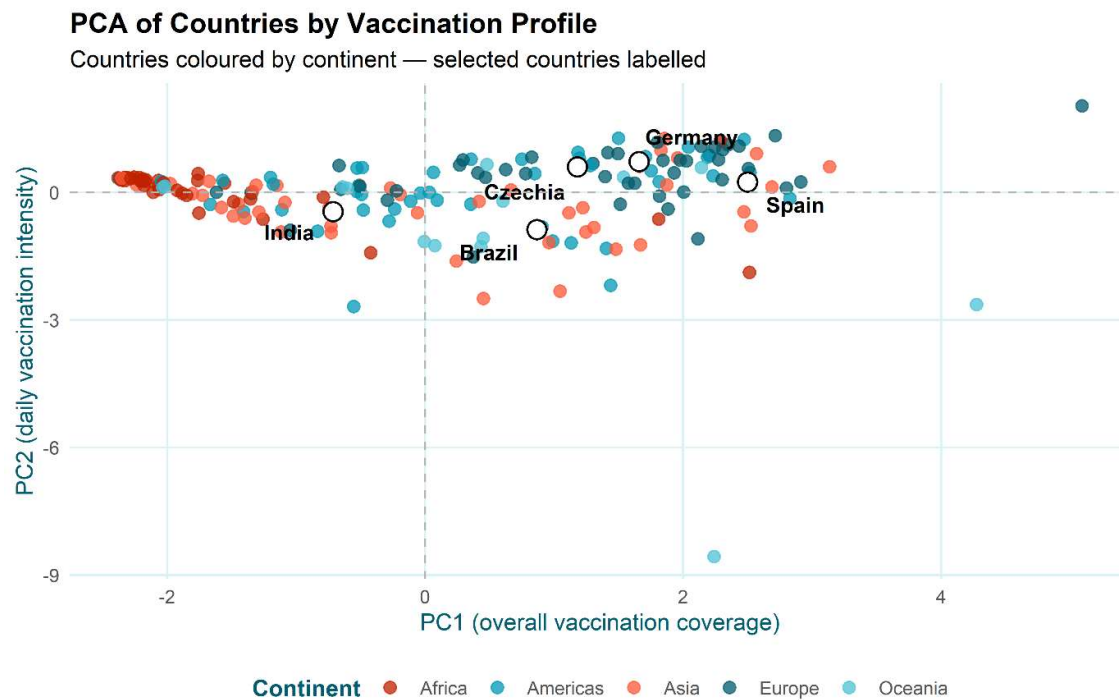


Figure 14. PCA of countries by vaccination profile, coloured by continent.

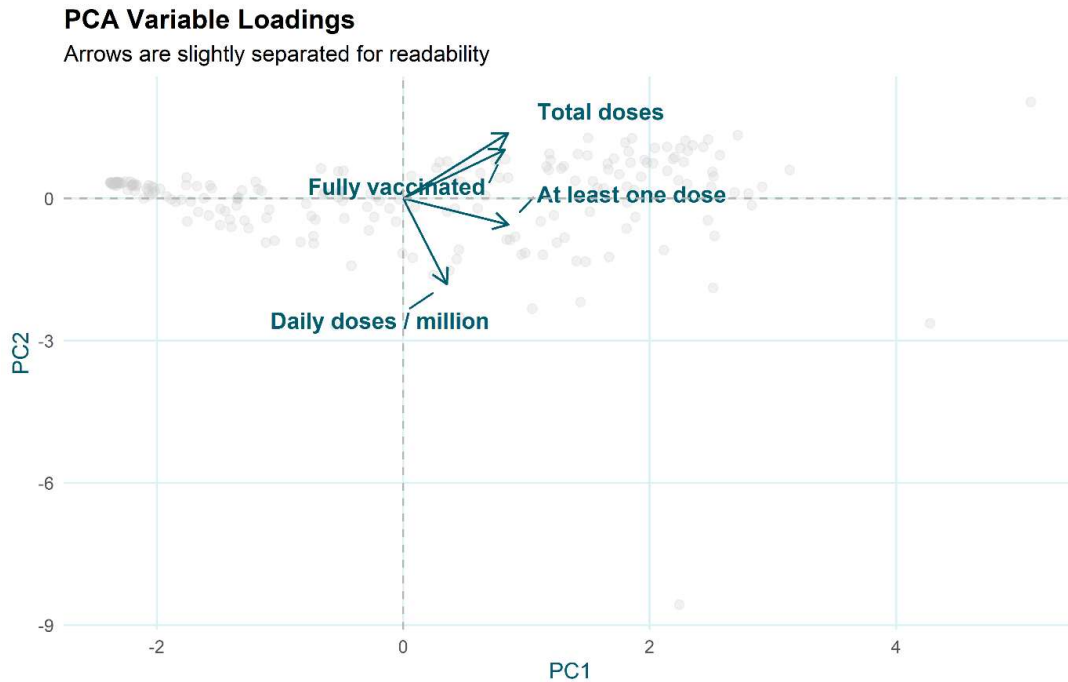


Figure 15. PCA variable loadings.

Before fixing the number of clusters, both the elbow method and the average silhouette width were computed (Figure 16); together with the theoretically expected high/moderate/low segmentation, they support retaining $k = 3$.

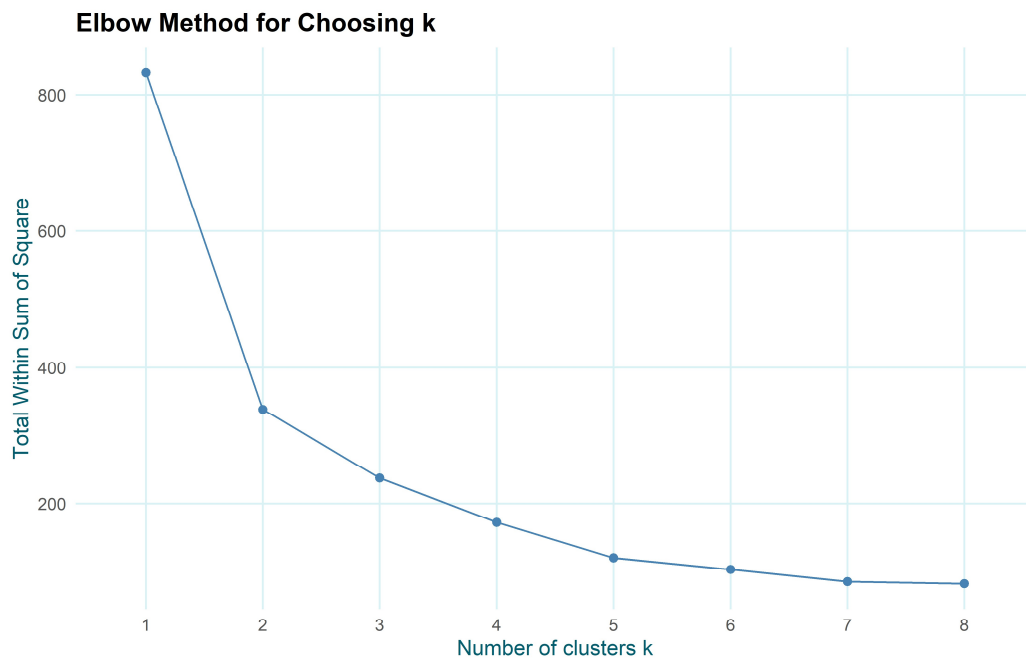


Figure 16a. Elbow method (within-cluster sum of squares) for choosing k .

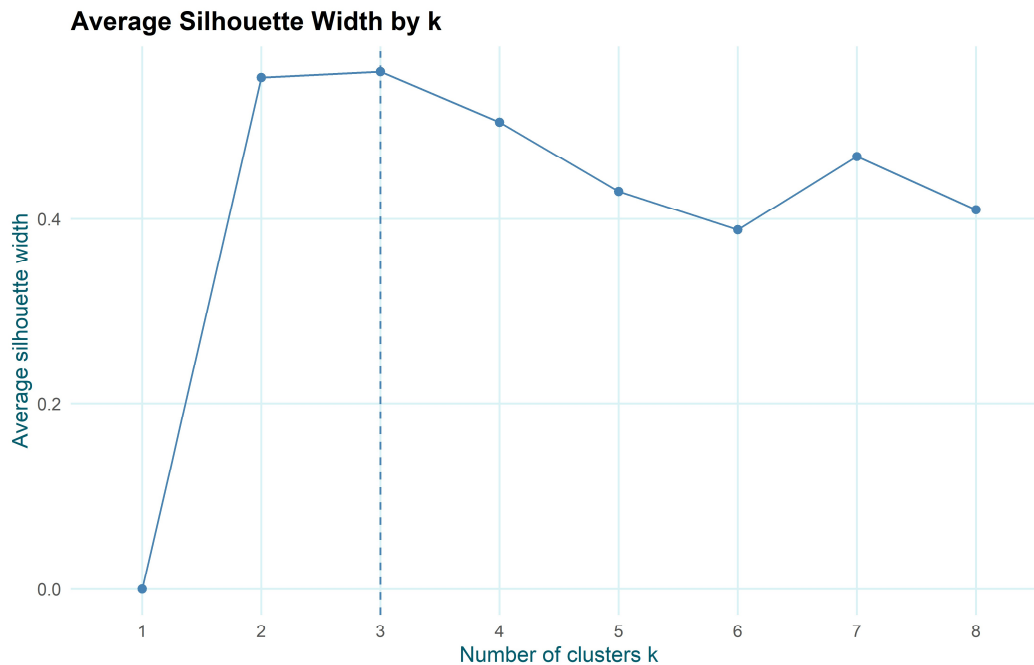


Figure 16b. Average silhouette width by k.

K-means clustering (Figure 17) identifies three structurally distinct groups, labelled from each cluster's own computed mean coverage: a high-coverage profile of 70 countries (mean 59.7 fully vaccinated per 100), a moderate-coverage profile of 35 countries (mean 32.7), and a low-coverage profile of 104 countries (mean 8.9).

Spain, Germany, and Czechia fall in the high-coverage cluster; Brazil in the moderate-coverage cluster; India in the low-coverage cluster consistent with the trajectories already shown in Figure 1.

K-means Clusters in PCA Space (k = 3)

Cluster labels assigned from each cluster's own computed mean coverage

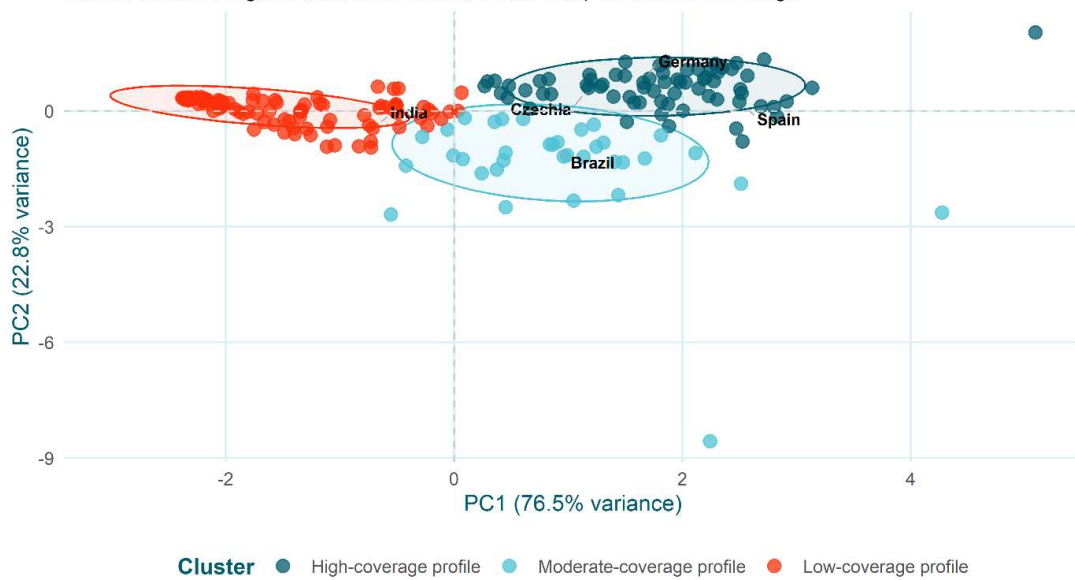


Figure 17. K-means clusters in PCA space (k = 3); labels assigned from each cluster's own computed mean coverage.

5.8 ARIMA Forecasting

Germany's daily coverage series (gaps interpolated) was modelled with `auto.arima()`, which selected `ARIMA(2,2,1)(2,1,0)[7]`; residual diagnostics showed no significant leftover autocorrelation (Ljung-Box $Q^* = 13.72$, $df = 9$, $p = 0.133$). On an 80/20 train-test split, ARIMA outperformed an ETS benchmark (RMSE 2.44 vs. 3.05; MAE 1.69 vs. 2.15), supporting it as the better-suited model for this series.

ARIMA Forecast: Germany Full Vaccination Coverage

30-day forecast with 80% and 95% confidence intervals

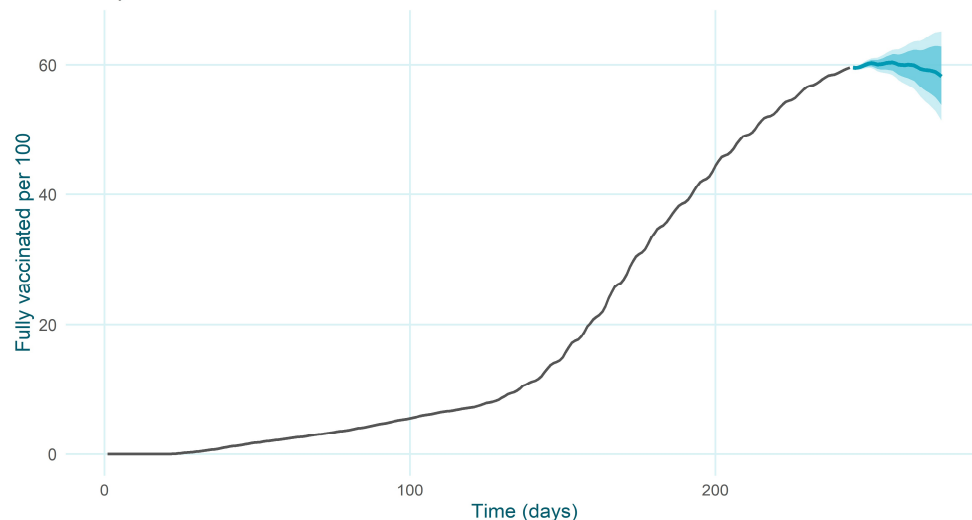


Figure 18. ARIMA forecast of Germany's full vaccination coverage (30-day horizon, 80% and 95% prediction intervals).

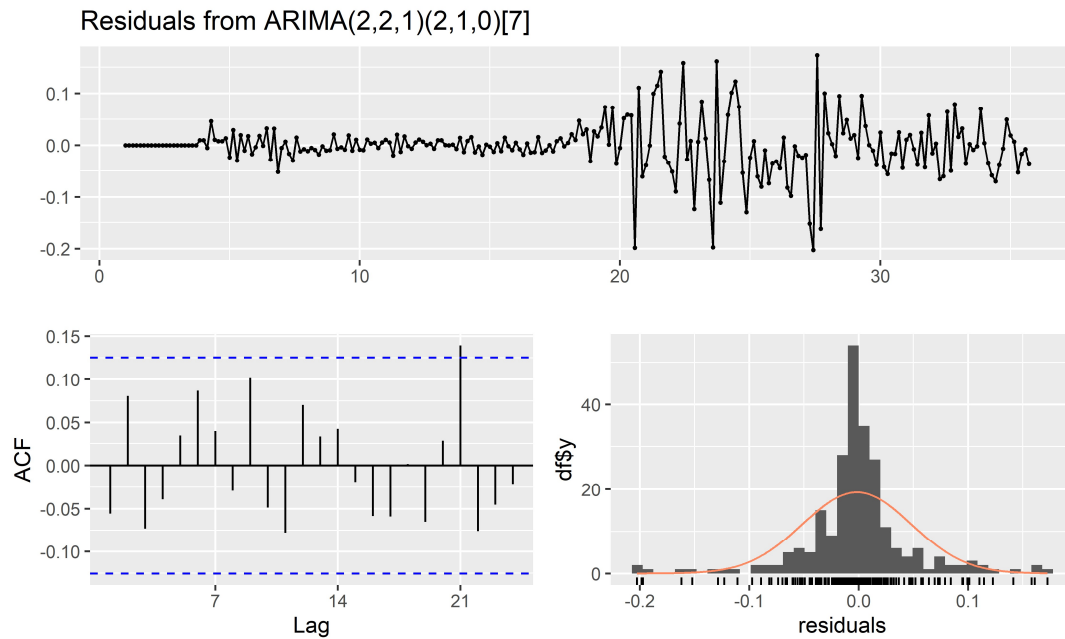


Figure 19. Residual diagnostics for Germany's ARIMA model.

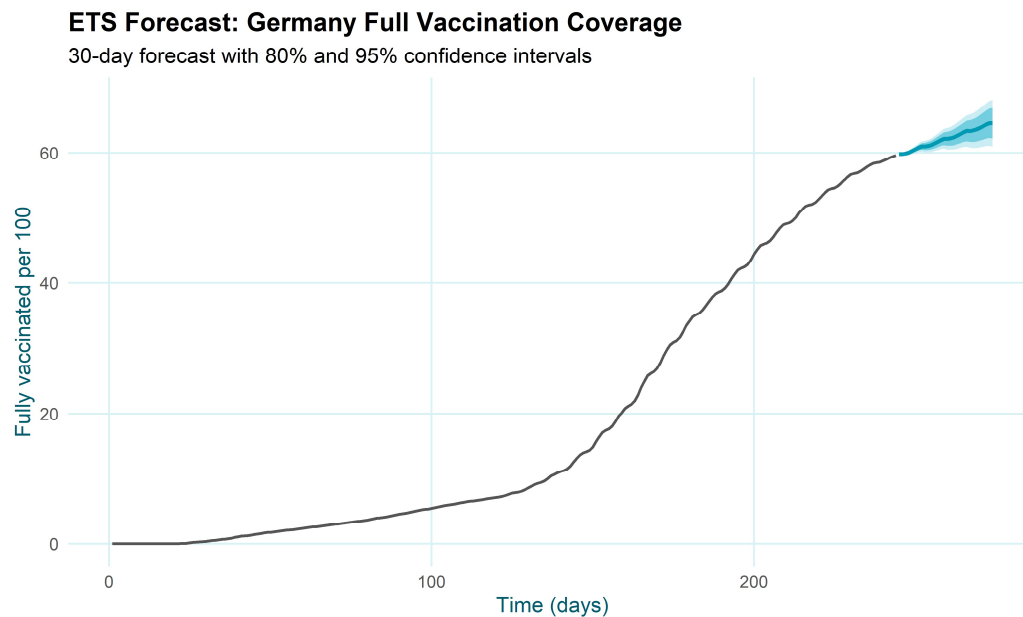


Figure 20. ETS forecast of Germany's full vaccination coverage, shown for comparison.

The same modelling strategy auto-selected differencing applied to a gap-filled level series, with no manual pre-differencing was then extended to Spain, India, and Bangladesh (Figure 21). Spain's model selected order (1,1,1)(1,1,1)[7] with an Augmented Dickey-Fuller p-value of 0.559 on levels (consistent with a unit root, supporting differencing); India selected (0,2,2)(2,0,0)[7] ($p = 0.990$); Bangladesh selected (0,2,3)(2,0,0)[7] ($p = 0.951$).

ARIMA Forecast of Full Vaccination Coverage: Spain

30-day forecast with 80% and 95% confidence intervals

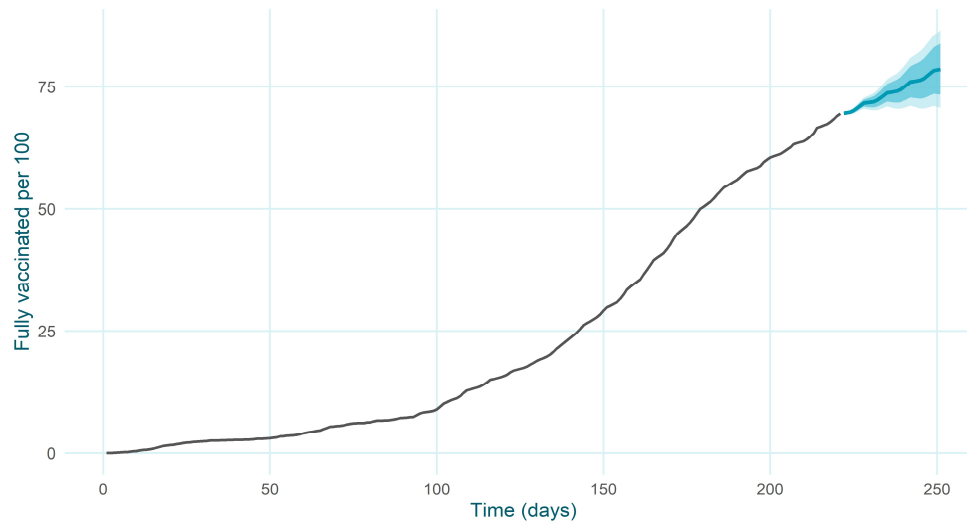


Figure 21a. ARIMA forecast of full vaccination coverage: Spain.

ARIMA Forecast of Full Vaccination Coverage: India

30-day forecast with 80% and 95% confidence intervals

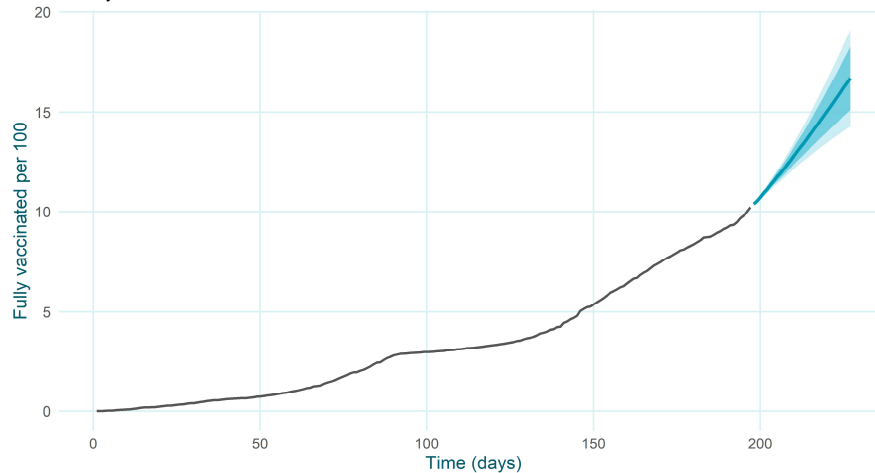


Figure 21b. ARIMA forecast of full vaccination coverage: India.

ARIMA Forecast of Full Vaccination Coverage: Bangladesh

30-day forecast with 80% and 95% confidence intervals

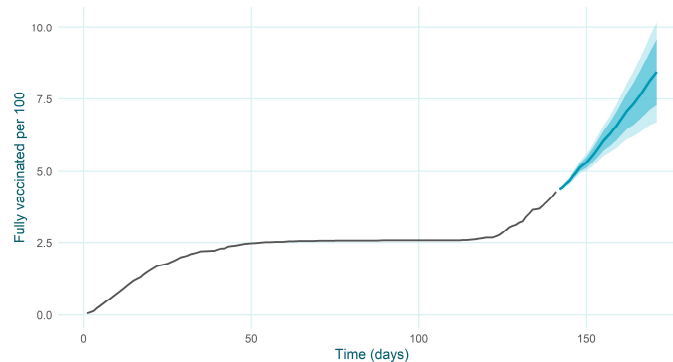


Figure 21c. ARIMA forecast of full vaccination coverage: Bangladesh.

The two uncertainty metrics in Figure 22 tell different stories. In absolute terms (22a), Spain's 30-day forecast had the widest 95% interval of the three countries (7.04 percentage points, vs. 1.99 for India and 1.41 for Bangladesh) the opposite of what was expected if higher income meant more reliable forecasts. This is a scale effect: Spain's model needed full seasonal differencing (compounding uncertainty faster over multiple weeks), and its coverage was already near saturation (~70%), where absolute swings simply have more room to occur than in a series still rising from near zero.

Rescaled by each country's own forecast level (22b), the ranking flips: Bangladesh's forecast is proportionally most uncertain (22.4% of its forecast level), then India (14.9%), then Spain (9.5%). Absolute and relative uncertainty answer different questions, how wide the band is in real terms, versus how uncertain the forecast is relative to what's being forecast and only the relative view supports the idea that forecast reliability tracks income level once the scale confound is removed.

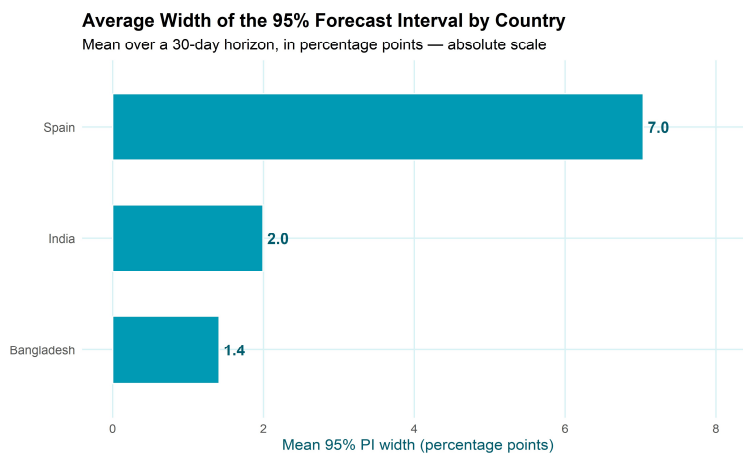


Figure 22a. Mean 95% prediction-interval width by country, in absolute percentage points.

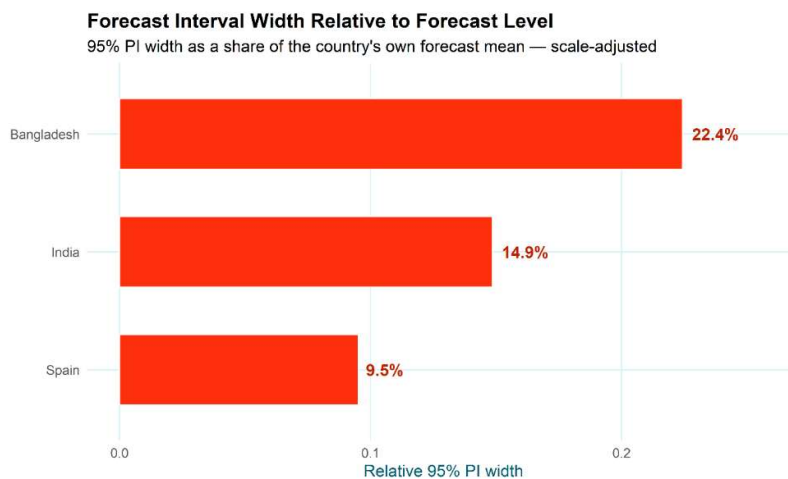


Figure 22b. Mean 95% prediction-interval width by country, relative to each country's own forecast level.

6. Discussion

6.1 Strengths

This analysis combines descriptive statistics, unsupervised learning, and time series forecasting to characterize COVID-19 vaccination inequality from cross-sectional, longitudinal, and predictive angles, using a single, regularly updated public dataset. World Bank income-group aggregates already embedded in the data were used directly, avoiding an external country-to-income crosswalk. The ARIMA component adds a forward-looking dimension: by testing rather than assuming whether forecast uncertainty differs by income level, and reporting both the absolute result (which contradicted the initial hypothesis) and the scale-adjusted result (which supported it), the analysis shows that inequality can be structural to the data-generating process itself, not just to the outcomes it measures.

6.2 Limitations

Several limitations apply. First, cross-sectional comparisons based on each country's "latest observation" (Figures 2, 3, 12, 14, 17) aren't all from the same date, they range from April to August 2021, and at least 15 of the 216 countries have snapshots over 30 days stale, so their true position by late August could be higher than shown.

Second, as Section 5.3 shows, near-complete missingness doesn't mean no vaccination activity, so the coverage gap reported here may itself understate reality for the most data-poor countries.

Third, the observation window ends in August 2021, missing the substantial 2022–2023 expansion of rollout in lower-income countries.

Fourth, PCA excluded seven countries (3%) with incomplete indicator data, a small but non-random subset.

Finally, the ARIMA uncertainty comparison in Section 5.8 illustrates a broader caution: absolute and scale-adjusted metrics can point in opposite directions, and neither should be reported alone when countries differ as much in baseline coverage as Spain, India, and Bangladesh do.

6.3 Implications

These findings carry direct implications for public health policy and governance. The concentration of early vaccine access in high-income countries, combined with systematic under-reporting from lower-income ones, points to a need for both more equitable distribution frameworks

and stronger investment in health-data infrastructure. The forecast-reliability finding adds a specific implication: agencies using short-term forecasting for vaccine allocation or logistics should expect and plan for wider relative uncertainty in exactly the countries that need planning support most, rather than treating that imprecision as a neutral gap.

7. Product Proposal

7.1 Market Opportunity

The findings above describe a monitoring failure as much as a vaccination failure. Most of the world never reached the 50% threshold; speed and coverage are distinct problems; 36 countries are effectively invisible in the data; and forecasts are least reliable precisely where planning support is scarcest. The institutions that allocate vaccine funding: the WHO, GAVI, UNICEF, the World Bank, and large non-governmental organisations must decide against exactly this picture, and no single instrument presents it to them. This is an unmet need the present analysis is positioned to address.

7.2 Product Concept

Our product is proposed as a health-equity monitoring service: a dashboard and alerting layer tracking, in near-real time, which countries are falling behind, how quickly they are progressing, and how far their reported data can be relied upon. It would ingest the daily Our World in Data feed and return a ranked, confidence-weighted view of where additional funding would avert the most harm. Each module derives directly from a finding in this report:

- ☀ 71.8% of countries never reached 50%; coverage spans an 11,640-fold range (5.1)
 - **Coverage tracker:** the addressable population is most countries, not a narrow segment.
- ☀ Time-to-threshold ranged 0–235 days, independent of coverage level (5.2)
 - **Rollout-speed indicator:** speed is a leading signal, coverage a lagging one a stalling campaign shows up months before coverage reflects it.
- ☀ 36 countries ≥95% missing; Tokelau 100% missing yet ~96% vaccinated (5.3)
 - **Data-gap flagging:** missing data gets its own risk state unmonitored, not unvaccinated.
- ☀ Daily throughput correlates only weakly with cumulative coverage (5.6)
 - **Ranking logic:** rankings use cumulative position and trajectory, not the latest daily figure.
- ☀ PCA/k-means yield three interpretable country profiles (5.7)
 - **Triage segmentation:** a prioritised shortlist instead of an undifferentiated 200-country table.
- ☀ Relative forecast uncertainty is 2.4× higher in Bangladesh than Spain (5.8)
 - **Confidence-aware forecasting:** every projection ships with its own reliability score.

7.3 Feasibility and Recommendation

The underlying data are public and freely available, so acquisition costs are negligible. The commercial value lies in the analytical layer segmentation, data-reliability scoring, and uncertainty quantification for which this report serves as a working prototype. The team's competence in statistical modelling is the principal barrier to entry: the confidence-aware forecasting component cannot be reconstructed from the raw feed alone.

Two limitations must be stated plainly. The prototype relies on a static 2021 snapshot, and the ARIMA specification requires the structural correction described in Section 6.2 before any forecast could be released to a client. Neither argues against proceeding; both argue for a limited pilot rather than immediate development at scale.

It is therefore recommended that leadership approve a three-month pilot developing the coverage, speed, and data-reliability modules, validated against the 2021 period analysed here, with the objective of securing two design-partner institutions before committing to full-scale development.

8. Conclusions

COVID-19 vaccination trends differed substantially across countries between December 2020 and August 2021. High-income countries reached higher coverage earlier and more consistently; only 28.2% of countries reached the 50% threshold at all, and among those, time-to-threshold ranged from 0 to 235 days. Missing-data patterns, the PCA/clustering structure, and the ARIMA uncertainty comparison all point the same way: inequality in this period was structural, extending beyond vaccination outcomes into the reporting infrastructure and predictive tools used to monitor them. Worth restating plainly: this last claim holds only once forecast uncertainty is scaled to each country's own coverage level the naive, unscaled comparison pointed the other way, and reporting that inversion honestly is part of the conclusion, not a complication to hide.

Future work could extend this analysis through 2022–2023 to capture later rollout phases, once comparably clean data exist. Adding covariates such as GDP per capita, healthcare expenditure, COVAX participation which would allow a more granular look at what drives vaccine access.

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Appendix

Appendix A: Dataset Variable Descriptions

The following variables were used in this analysis:

- 🌻 **people_fully_vaccinated_per_hundred**: share of the population that received all doses prescribed by the original vaccination protocol, per 100 inhabitants.
- 🌻 **total_vaccinations_per_hundred**: total number of doses administered, per 100 people.
- 🌻 **daily_vaccinations_per_million**: daily doses administered per million population (7-day rolling average).
- 🌻 **location, date, iso_code**: identify the reporting unit (country, territory, continent, or income group) and the calendar date of each observation.

Appendix B: Software and Reproducibility

All analysis was conducted in R using `dplyr`, `tidyr`, `ggplot2`, `forecast`, `tseries`, `zoo`, `factoextra`, `sf`, `rnaturalearth`, `countrycode`, `ggcorrplot`, `ggdist`, `ggbeeswarm`, `ggrepel`, and `ggbreak` packages. A fixed random seed (42) was set prior to k-means fitting.

Appendix C: Companion R Notebooks

The complete, runnable R Markdown notebooks underlying this report `vaccination_pipeline.Rmd` (descriptive statistics, mapping, PCA, and clustering) and `arma.Rmd` (time series forecasting) are submitted as separate files alongside this report.

[Github link](#)